Caching and stale information

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

Ask which check decides this request before discussing speed.

Teach current permission, context keys and strict TTL boundaries with checked fixtures and honest publication limits.

Keep one habit: Model the three checks → contrast young misses with expiry → inspect approved-version lookup → assess independent reasoning → record limits.

MAIN LESSON · SECTION 1

Prepare the connected service investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of caching and stale information

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Data retention and deletion. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of data retention and deletion

Reference example, not the only acceptable wording: Teacher explanation: Retention and deletion cover all declared stores, including caches and logs. Removing a visible row may leave copies elsewhere.

Toy policy retains session content for 7 days and deletes it when age is at least 7. Audit metadata may remain for 30 days without content. Track primary storage, cache and backup policy separately.

Without opening the old lesson, explain one key idea from Untrusted **input** boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of untrusted **input** boundaries

Reference example, not the only acceptable wording: Teacher explanation: Untrusted **input** must be treated according to a contract before reaching code, queries or tools.

Data that resembles instructions does not acquire execution authority. For the toy lookup accept room IDs matching one uppercase letter plus one digit. Perform exact dictionary lookup; never evaluate user strings as code.

Reject invalid length or characters.

Without opening the old lesson, explain one key idea from Approval gates and idempotency. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of approval gates and idempotency

Reference example, not the only acceptable wording: Teacher explanation: Approval gates bind side effects to an actor, exact version and destination.

Idempotency prevents repeated requests with the same key from repeating a committed effect. The mock submission requires approval for request version v1 and destination D. Store key -> committed result.

Reusing the same key with the same payload returns that result; the same key with different payload is rejected.

Pictured end goal

Explain caching and stale information using the deciding step in this fictional service.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Data retention and deletion

Untrusted input boundaries

Approval gates and idempotency

Use the three-session teaching cycle above, with extra time as needed. Caching reuses results but needs freshness and access-aware keys. A cached answer must not cross role or corpus-version boundaries. Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions. Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of caching and stale information

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach cache reuse as three independent requirements

> Keep current authority, matching evidence context and a strict age boundary visible.

Definitions and complete classroom contract

Caching reuses results but needs freshness and access-aware keys. A cached answer must not cross role or corpus-version boundaries.

Cache key includes query, role, corpus version and procedure version. Entry valid only while age<TTL. At exact TTL it expires. Teacher and student results never share a role-agnostic key.

Check	Complete classroom rule
Current permission	Trusted Boolean fixture checked before any cached value is returned
Cache key	Query, trusted role, corpus version and procedure version
Trusted role	Student or teacher supplied by the harness; not a request-body claim
Age and TTL units	Virtual minutes; age = now minus saved time
Freshness	0 ≤ age < TTL; exact equality expires
Missing or changed context	Miss: do not return a differently keyed entry
Miss is not an answer	Only an authorized lookup of approved evidence can supply a replacement
Scope	Role-only separation is sufficient only for these fixtures; real user, tenant or object differences need appropriate additional scope
What TTL cannot prove	Truth, current permission, correct publication versions or physical deletion

Main decision ledger and checked reasoning

Main fixture: TTL 5 minutes	Request at virtual minute	Expected result and reason
Teacher c1/p1 entry saved at 0; teacher c1/p1 allowed	4	Hit; matching key and age 4 < 5
Same query, student c1/p1 allowed	4	Miss; teacher entry has a different role key
Teacher c1/p1 permission revoked	4	Denied before cached content is returned
Teacher c1/p1 allowed	5	Expired miss; age equals TTL
Teacher c2/p1 allowed	1	Miss; corpus version changed despite young age
Teacher c1/p2 allowed	1	Miss; procedure version changed despite young age

Check permission before serving any value. In the main TTL 5 fixture, age 4 teacher c1/p1 hits, student c1/p1 misses, revoked permission denies, age 5 expires, and corpus/procedure changes miss at age 1. The separate picture holds role/procedure fixed: current version 5 must not reuse version 4's 09:00; lookup version 5's 09:30 and cache it under its own key.

A later hit still needs current permission and age below TTL. The code uses trusted fixtures, not authentication. Same-role users may require further user/tenant/object scope in real systems. Silent unversioned edits remain undetected, and unserved entries are not automatically deleted. Reserve the selected TTL 10 version-change reasoning; the old compact helper already demonstrates equality expiry.

Caching and stale information — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Teacher response is cached by query only and returned to a student.	Cache key includes query, role, corpus version and procedure version. Entry valid only while age<TTL. At exact TTL it expires. Teacher and student results never share a role-agnostic key.	Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

Qualification: A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Teacher response is cached by query only and returned to a student.

Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Boundary and publication probes

Probe	Expected interpretation
Age just below TTL with matching context and permission	May serve the saved authorized value; no new truth guarantee
Age exactly TTL or above	Miss; expired
Negative age	Miss; inconsistent supplied timestamps
NaN time or nonpositive TTL	Reject invalid fixture
Permission changes after saving	Denied even while entry is fresh
Role changes but query stays the same	Different key; never reuse the teacher-only value for a student
Corpus or procedure version changes	Different key; young old-version entry cannot satisfy it
Source changes without changing version	May go undetected; publication process must maintain versions
Expired entry still exists in dictionary	Not served, but not deleted; retention is a separate policy

Separate opening-time case and fixed-context qualification

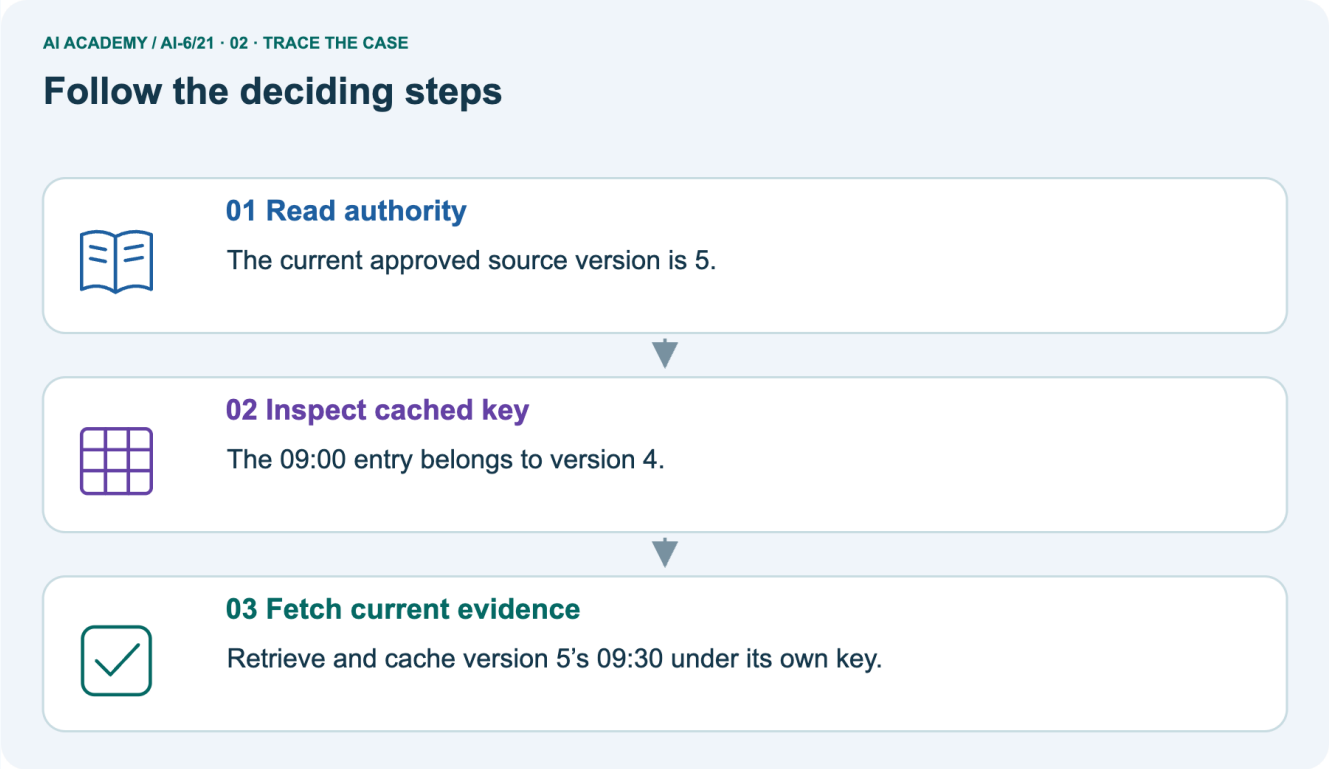
Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved.

Version 4 says 09:00; version 5 says 09:30. **Cache key** contains room and source version.

The picture simplifies the key to room and source version while holding the trusted role and procedure fixed. Approved version 5 says 09:30; the saved version 4 entry says 09:00. The first version 5 request must miss that old key even if the entry is young. Its authorized local fixture lookup supplies 09:30 and saves a separate version 5 entry.

A later request may hit that entry only while permission still applies and its age remains below TTL. Reliable publication must change the version when evidence changes; a silent edit under the same version is invisible to these key and age checks.

Picture trace and later-hit conditions



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read authority	The current approved source version is 5.
2. Inspect cached key	The 09:00 entry belongs to version 4.
3. Fetch current evidence	Retrieve and cache version 5's 09:30 under its own key.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A cache saves repeated work but can retain obsolete facts. Version-aware keys distinguish answers grounded in different records.

Version keys cannot detect silent unversioned edits; the source publication **process** must update versions reliably.

Change: The next request still uses approved version 5.

Prediction: The matching version 5 entry can supply 09:30 under the freshness policy.

Reason: Request and cached source version now agree.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Cache key	Fields identifying a reusable result
TTL	The lifetime allowed by the cache contract
Invalidation	Marking an entry unusable after a relevant change

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def cache_key(query, role, corpus_version, procedure_version):
    return (query, role, corpus_version, procedure_version)

def cache_valid(age, ttl, saved_version, current_version):
    return 0 <= age < ttl and saved_version == current_version

print(cache_valid(10, 10, "v1", "v1"))
```

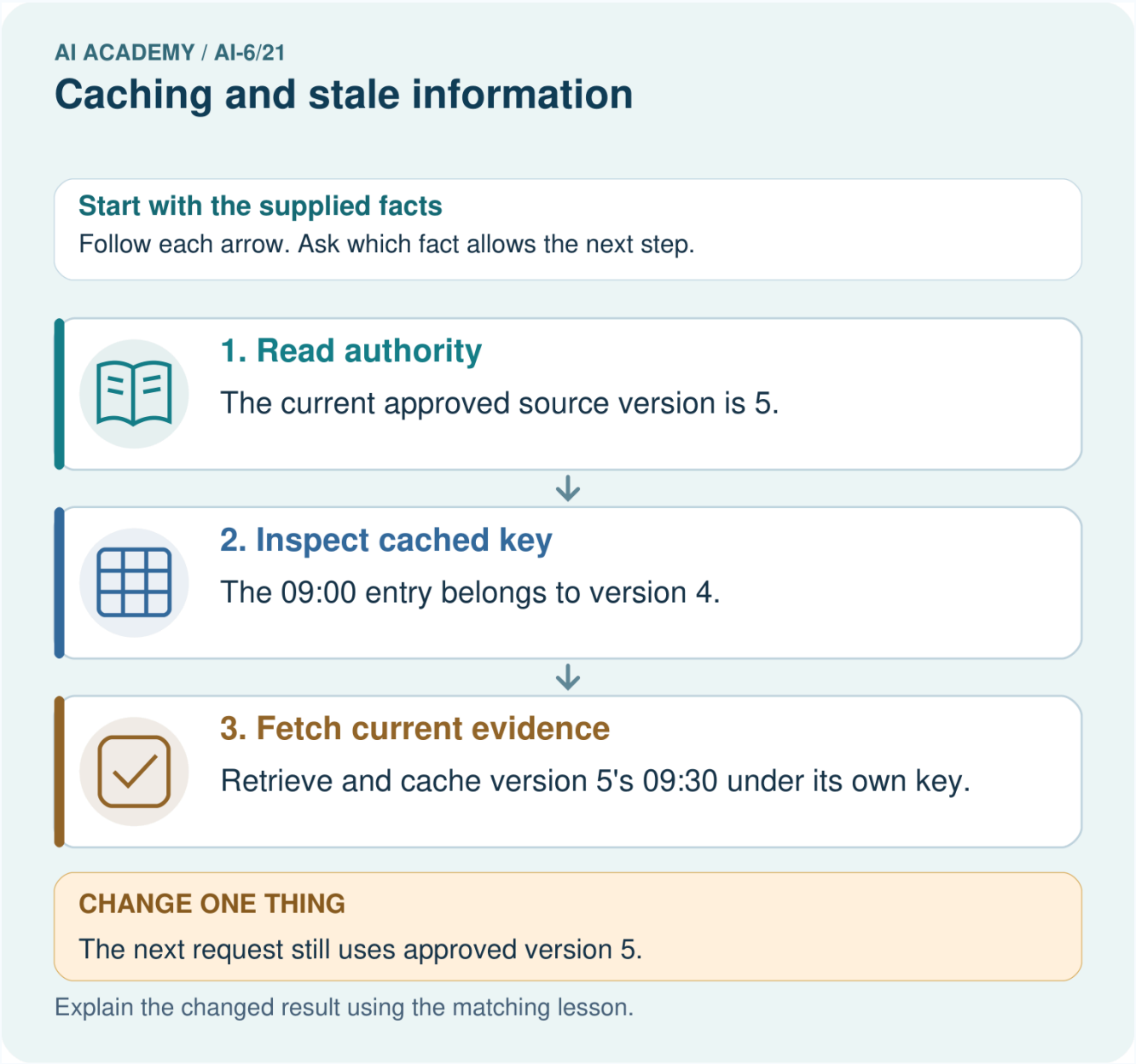
Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move	Teacher prompt
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Caching and stale information

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Teacher response is cached by query only and returned to a student. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Caching and stale information. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Identify the version mismatch in the cached 09:00 answer. What should the first version 5 request return, and what permits a later version 5 cache hit?

The current approved source version is 5. The 09:00 entry belongs to version 4. Retrieve and cache version 5's 09:30 under its own key. Changed case: The matching version 5 entry can supply 09:30 under the freshness policy. Request and cached source version now agree.

Changed-case reasoning: The matching version 5 entry can supply 09:30 under the freshness policy. Request and cached source version now agree.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain section 1 and vocabulary. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main teacher age 4 hits only with matching context/current permission; student role misses, revoked permission denies, exact age 5 expires and changed corpus/procedure misses. Picture first version 5 request misses the version 4 key and looks up 09:30; later version 5 can hit while fresh and authorized. Selected independent TTL 10: age 9 is within time but old corpus version is invalid; ages 10 and 11 expire as well.

State both version and time conditions. The compact prior helper demonstrates equality expiry, so do not count that familiar boundary alone as unseen transfer.

Diagnose the current misconceptions

The worked result establishes every future case: A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new service scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: TTL 10 minutes; ages 9, 10, 11. Corpus version changes at age 9. Decide validity.

Teacher explanation: Age 9 would fit time but version change invalidates it. Ages 10 and 11 expire. Both freshness and version conditions must hold.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: Caching reuses results but needs freshness and access-aware keys. A cached answer must not cross role or corpus-version boundaries. **Cache key** includes query, role, corpus version and procedure version.

Entry valid only while age<TTL. At exact TTL it expires. Teacher and student results never share a role-agnostic key.

W2 Trace the worked case

Teacher response is cached by query only and returned to a student. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

W3 Audit the limit

A report omits this qualification: A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

Qualification: A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

TTL 10 minutes; ages 9, 10, 11. Corpus version changes at age 9. Decide validity.

Teacher explanation: Age 9 would fit time but version change invalidates it. Ages 10 and 11 expire. Both freshness and version conditions must hold.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Age 9 would fit time but version change invalidates it. Ages 10 and 11 expire. Both freshness and version conditions must hold. Require intermediate reasoning and the stated scope.

Fresh retry: TTL 5; age 4 with same role/version and age 5. Expected reasoning: Age 4 valid if permissions still apply; age 5 expired under strict age<TTL.

If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit service design

> Use saved evidence and boundary decisions to choose the next step.

Project and next-week bridge

Add a labelled caching and stale information **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of caching and stale information.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Queues and asynchronous work. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Delayed return and support record

Suggested return points in this level: ai-6/22, ai-6/24, ai-6/27. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of caching and stale information

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Teacher response is cached by query only and returned to a student. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

Check every learner before discussion: Use the worked case for Caching and stale information. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A cached result retains the limitations of its original **evidence** and can become stale independently of TTL.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The next request still uses approved version 5. Require a **prediction** and a reason before comparing. Expected result: The matching version 5 entry can supply 09:30 under the freshness policy. Reason: Request and cached source version now agree.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

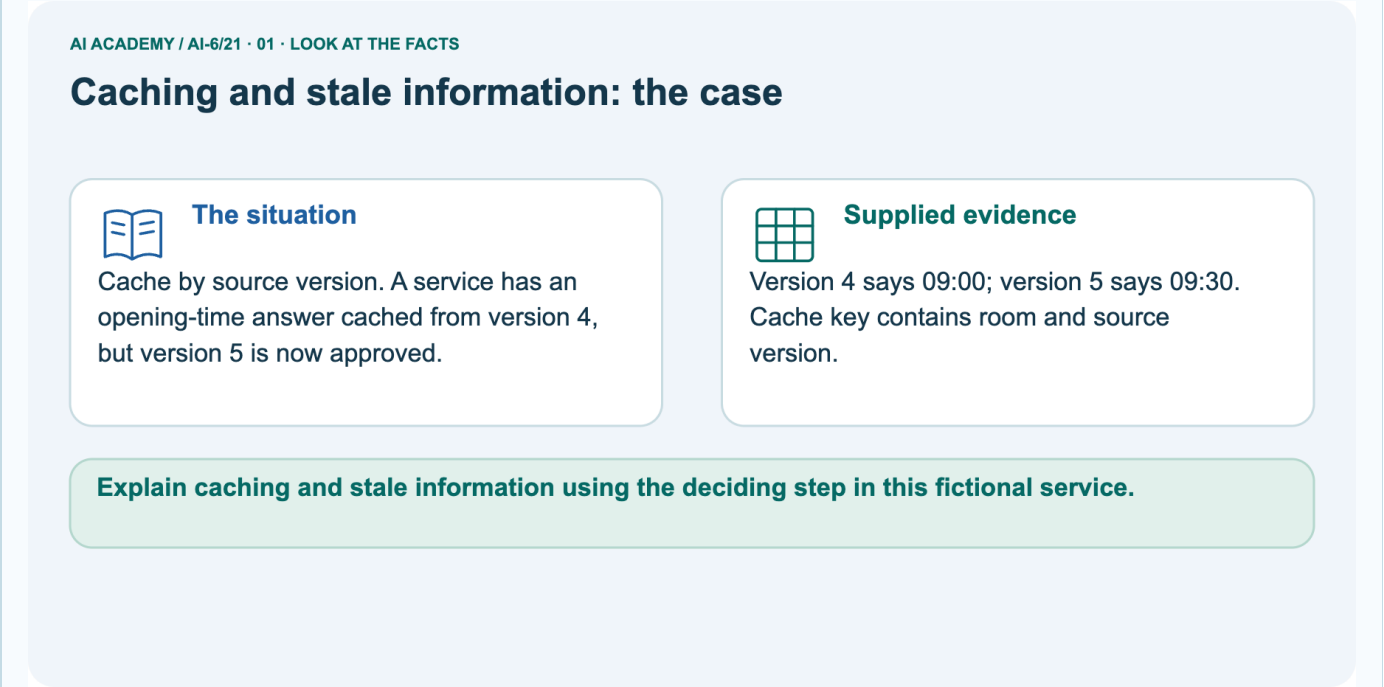
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Caching and stale information: the case



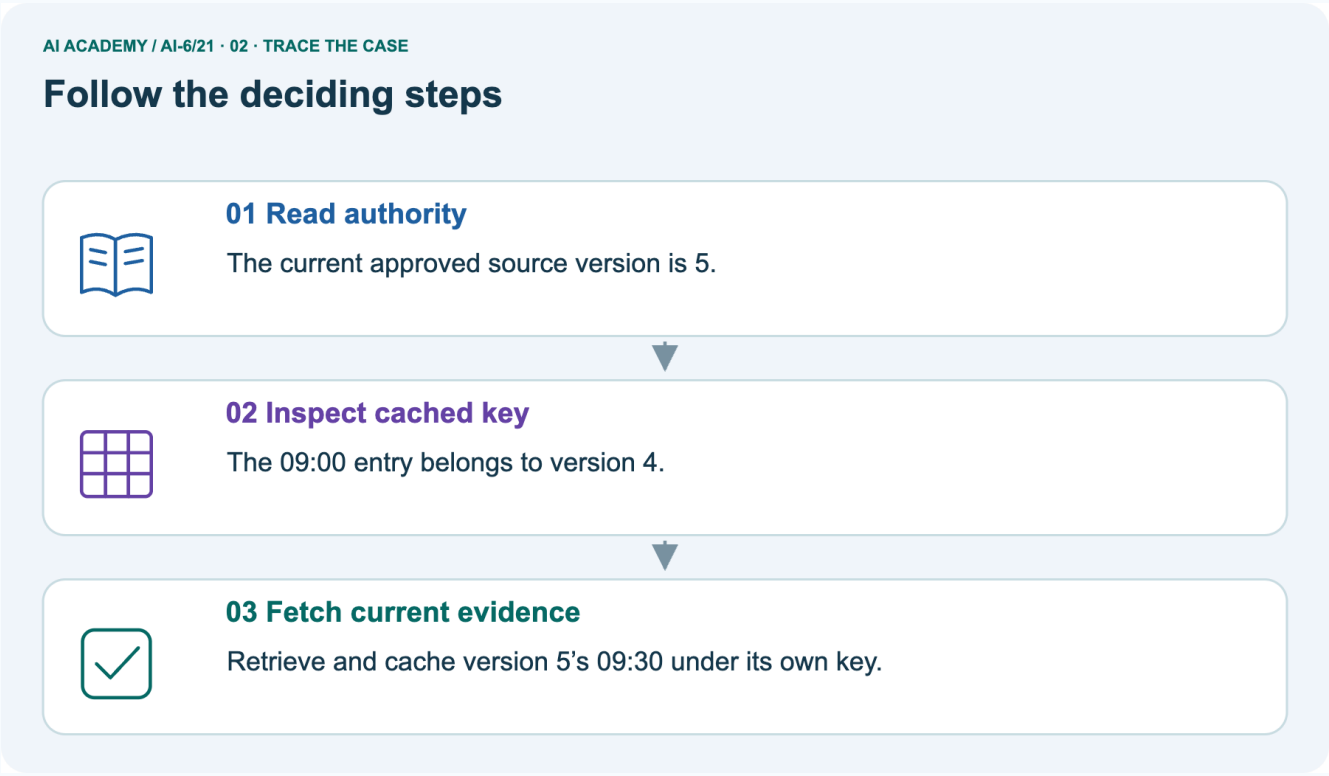
Follow the labelled picture with the matching explanation. Open the vector diagram

Cache by source version. A service has an opening-time answer cached from version 4, but version 5 is now approved. Supplied facts: Version 4 says 09:00; version 5 says 09:30. **Cache key** contains room and source version.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read authority: The current approved source version is 5. → Inspect cached key: The 09:00 entry belongs to version 4. → Fetch current **evidence**: Retrieve and cache version 5's 09:30 under its own key.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Caching and stale information — evidence trace

AI ACADEMY / AI-6/21 · 03 · INSPECT THE LESSON MODEL

Caching and stale information — evidence trace

Input and scope	Check or calculation	Conclusion and limit
Teacher response is cached by query only and returned to a student.	Cache key includes query, role, corpus version and procedure version. Entry valid only while age<TTL. At exact TTL it expires. Teacher and student results never share a role-agnostic key.	Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the labelled picture with the matching explanation. Open the vector diagram

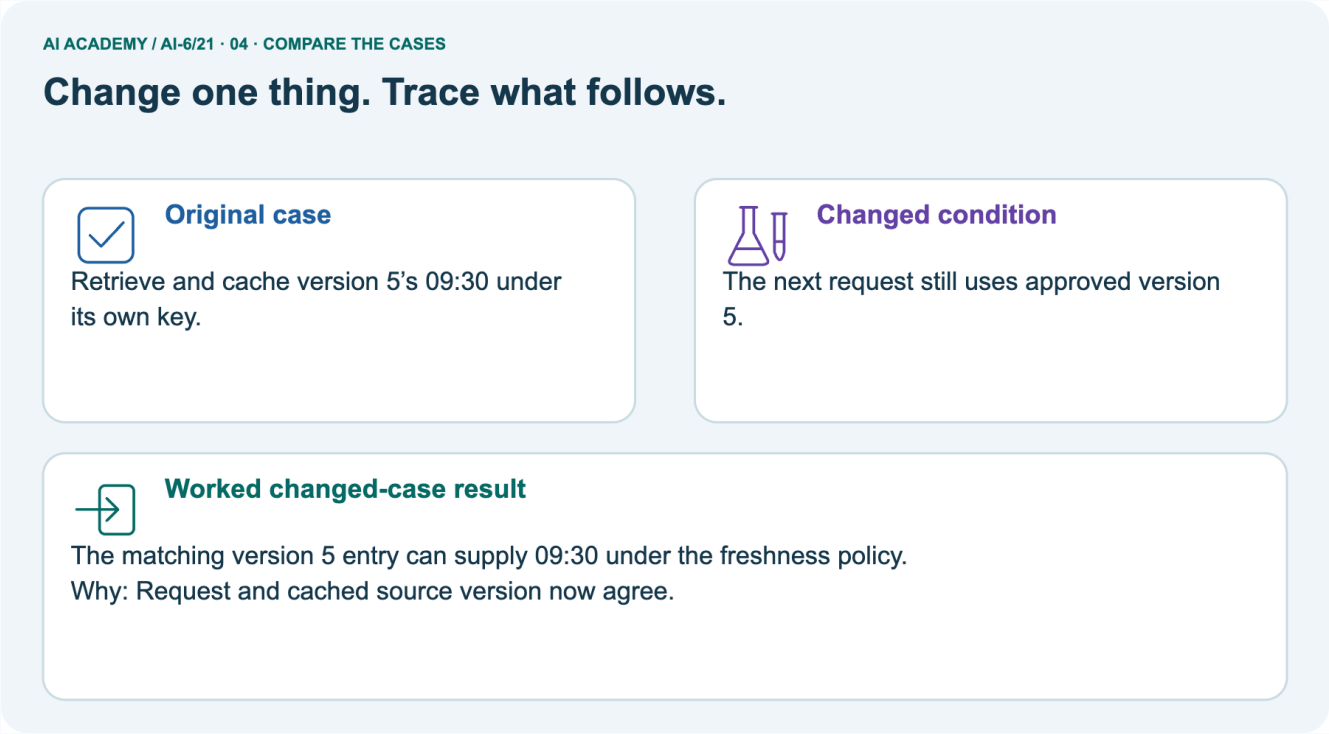
Caching and stale information — **evidence** trace. Teacher response is cached by query only and returned to a student. | **Cache key** includes query, role, corpus version and procedure version. Entry valid only while age<TTL.

At exact TTL it expires. Teacher and student results never share a role-agnostic key. | Role omission creates an access leak. Include role and enforce current permission before returning any cached content.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

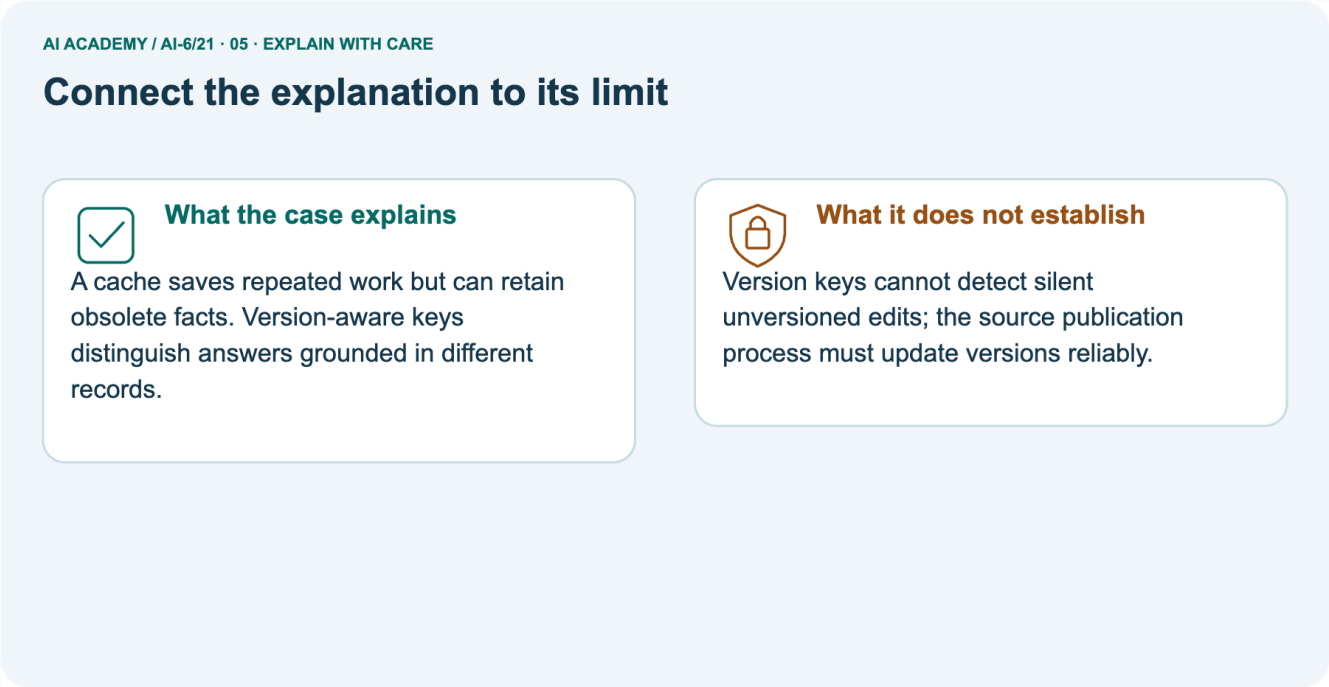
Original case: Retrieve and cache version 5's 09:30 under its own key. Changed condition: The next request still uses approved version 5. Expected result: The matching version 5 entry can supply 09:30 under the freshness policy. Reason: Request and cached source version now agree.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Request and cached source version now agree.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A cache saves repeated work but can retain obsolete facts. Version-aware keys distinguish answers grounded in different records.

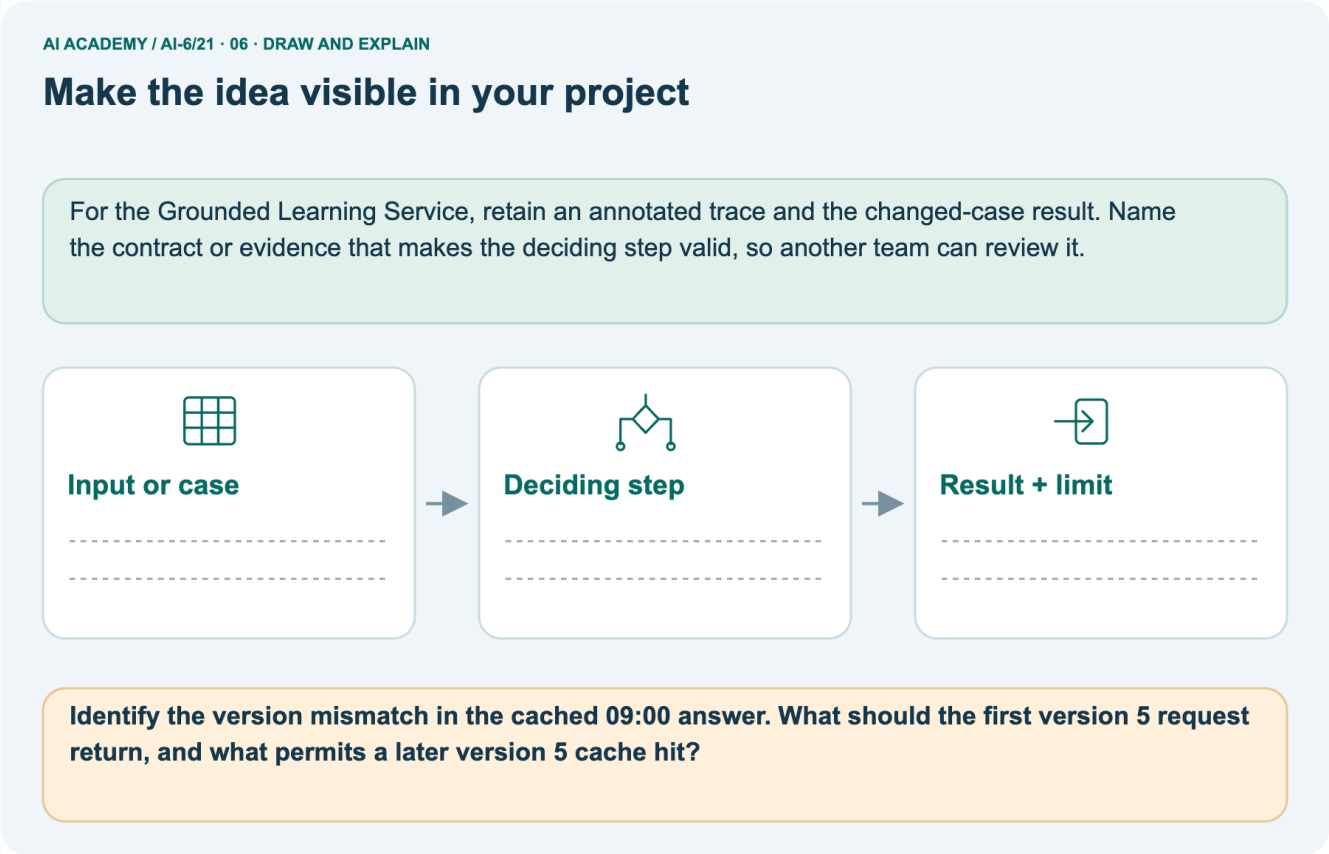
Limit: Version keys cannot detect silent unversioned edits; the source publication **process** must update versions reliably.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Version keys cannot detect silent unversioned edits; the source publication **process** must update versions reliably.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it. Draw your **input**, deciding step, result and limit. Supported question: Identify the version mismatch in the cached 09:00 answer. What should the first version 5 request return, and what permits a later version 5 cache hit?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see. Supported practice key: The current approved source version is 5. The 09:00 entry belongs to version 4. Retrieve and cache version 5's 09:30 under its own key. Changed case: The matching version 5 entry can supply 09:30 under the freshness policy. Request and cached source version now agree.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Caching and stale information: the case

02. Follow the deciding steps

03. Caching and stale information — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.